PRAVIN K



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CAREER OBJECTIVE

Electronics Engineering (VLSI) student with hands-on experience in Verilog and FPGA design on the Basys 3 board. I want to learn and contribute to hardware design, testing, and prototyping.

TECHNICAL SKILLS

Hardware Tools: Basys 3 FPGA Board
Software Tools: Xilinx Vivado, EDA Playground
Languages: Verilog HDL, SystemVerilog, C, C++ (Basic–Intermediate).
Core Knowledge Areas: Digital VLSI Design,FSM Design,PWM Generation.

EDUCATION

B.E. – Electronics Engineering (VLSI Design and Technology) 2027
Muthayammal Engineering College, Namakkal, Tamil Nadu
CGPA: 6.89 / 10.0

Higher Secondary Certificate – 12th Standard *2023*
Ananda Vidhyalaya Matric Higher Secondary School,Pattanam, Tamil Nadu
Percentage: 73.3%

Secondary School Certificate – 10th Standard *2021*
Ananda Vidhyalaya Matric Higher Secondary School,Pattanam, Tamil Nadu

MINI PROJECT

Smart Street Light Controller using Verilog HDL on Basys 3 FPGA Board

- Designed a 3-state FSM (DAY / NIGHT-DIM / NIGHT-ACTIVE) in Verilog HDL on the Basys 3 Artix-7 FPGA to control street light behaviour automatically. Built an 8-bit PWM generator to control LED brightness — 30% duty cycle in standby mode and 100% in active mode. Used onboard switches SW0 and SW1 to simulate LDR (light) and PIR (motion) sensor inputs, replicating real-world light and motion detection.
- Built a 27-bit clock divider and a 30-second motion timeout counter to handle timing, along with a 7-segment display controller to show the current FSM state in real time. Verified the full design using Xilinx Vivado simulation before deploying it to the board via USB-JTAG.
- Fixed a 7-segment display bug caused by incorrect anode/cathode signal mapping by checking the segment truth table in simulation. Also estimated a 70% reduction in energy consumption compared to a conventional always-on light, based on the standby vs active duty cycle difference.

INTERNSHIP EXPERIENCE

Production Intern – Wire Harnessing & NPD Department 2024-2025

Popular Systems, Coimbatore, Tamil Nadu

- Worked as a Production Intern in the Wire Harnessing and NPD department, gaining hands-on experience in wire harnessing assembly and end-to-end production workflows, including hardware component integration and prototype-level assembly. Learned manufacturing quality standards, component identification, and inspection procedures used in production-grade electronic assemblies, which is directly applicable to hardware testing and validation.
- Worked closely with cross-functional production teams to meet daily output targets while strictly following workplace safety protocols, gaining practical exposure to real-world manufacturing discipline and teamwork in a production environment.

ACHIEVEMENTS & CERTIFICATIONS

- **VLSI Design & Verification Training** - Attended a 19-day intensive RTL Design and Verification training program conducted by Taras Systems and Solutions, authorised by Muthayammal Engineering College. During this program, I gained hands-on experience in hardware design and functional verification using SystemVerilog and Universal Verification Methodology (UVM). I developed a foundational to intermediate level understanding of key verification components including testbench architecture, sequencer, driver, monitor, and scoreboard, along with practical simulation experience using EDA Playground.
- Workshop Certification – Lithium Battery Technology, conducted at the National Institute of Technology (NIT). Acquired in-depth knowledge of advanced energy storage systems, electrochemical principles, and battery management considerations, along with practical exposure to current trends in battery technology used in industrial and consumer applications.

LANGUAGES

Tamil &English

DECLARATION

I hereby declare that all the information furnished above is true and correct to the best of my knowledge and belief. I take full responsibility for the accuracy and authenticity of the details provided in this resume.

Place: RASIPURAM

Date: **[Pravin K]**